



Sensequake Larzé

Vibration Monitoring
System

USER MANUAL



LARZÉ

Sensequake

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QUICK START

This chapter outlines the necessary steps for taking vibration measurements with the Larzé Vibration Monitoring System. For comprehensive instructions, please consult the entire user manual.

1. **Setting Up the Sensors:** Start by connecting the antennas. Next, power on all the sensors, ensuring that each one's LED blinks green to indicate they are ready.
2. **Installing and Using the Control Application:** Install the Sensequake Larzé Control Application on your smartphone. Launch the app and tap on 'Menu' at the bottom. The app will scan for and list available Bluetooth sensors. Select sensor 1, labeled 'Manager' for multi-sensor setups. Alternatively, you can select any individual Bluetooth sensor for standalone operation.
3. **Sensor Network Configuration:** Navigate to the 'Sensors' tab to verify that all sensors are connected. Add sensors to the network by tapping the '+' icon and inputting their IDs. Available sensors will be shown in green boxes shortly.
4. **Initiating Measurement:** Select 'Start' from the bottom menu and enter your measurement settings. Proceed to the 'Main' tab, complete the test details, and press the play button to commence data acquisition.
5. **Observing Sensor Indicators:** Sensors will turn on a blue LED to indicate they have received the start command and are synchronizing. The blue LED will begin flashing after a short delay, signaling that data collection is underway. The Blue LED will stop when data acquisition is over.

6. **Retrieving Data:** To download the collected data, connect each sensor to a PC via the supplied USB cable.
7. **Recharging the Sensor:** Recharge the sensor battery by connecting it to a standard 5 V USB charger. The sensor can be powered on or off while charging.

Do not charge the sensors in temperatures below zero degrees Celsius.

LARZÉ PHONE APP

Sensequake Larzé Vibration System is configured using a phone app that can be installed on a device running an Android Operating System.

To get started, install the app on your smartphone. Then, make sure to enable Bluetooth access and location permissions for the app. The steps to do this may vary depending on your phone's operating system. For reference, Figure 2.1 illustrates an example of the permission settings on a phone.

The main screen, appearing after opening the Larzé app (Figure 2.2), consists of the following:

1. **Sensors:** to see a list of the expected sensors, their last status update and battery level.
2. **Start:** to adjust the sensor settings and start recording data.
3. **Plot:** to plot the last recorded file. You can also see real time vibrations on this page.
4. **Menu:** To establish a Bluetooth connection between a sensor and the phone, verify the radio signal strengths, see the latest logs or to characterize a geo-phone.

2.1 Menu

The menu consists of the following:

1. **Connect to Sensor:** To select a sensor from the list of Bluetooth devices. If Bluetooth is enabled on your phone, you can connect to a sensor from this

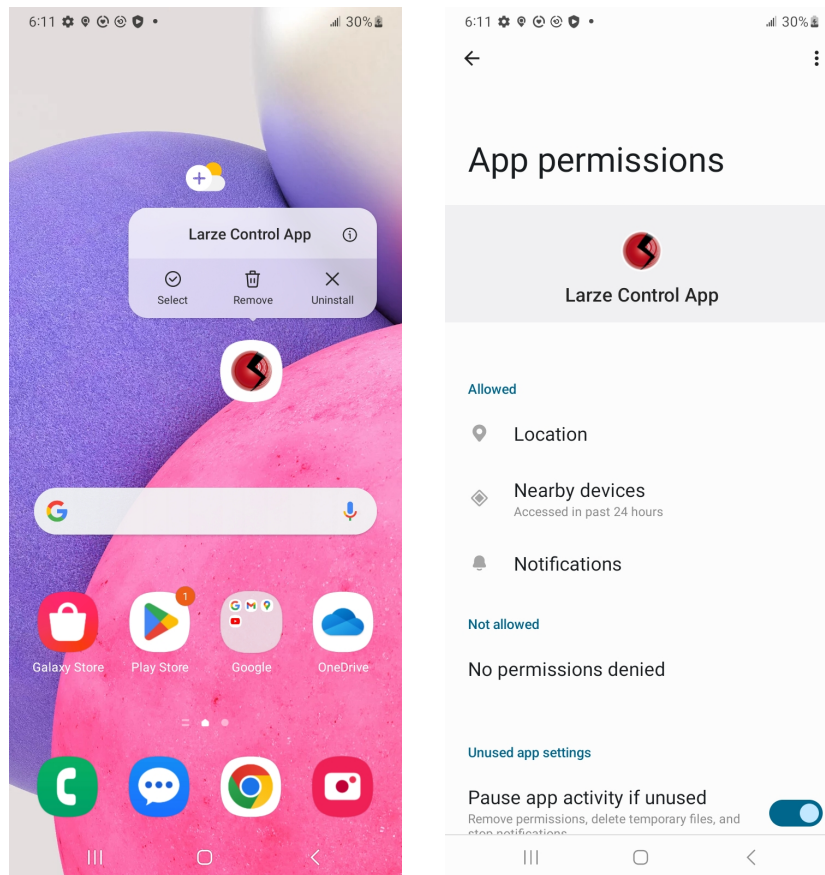


Figure 2.1: Enabling permissions for the app.

menu. Within a few seconds, a list of discovered Bluetooth sensors will appear. Sensor 1 (the manager) must have its Bluetooth turned on, while it is optional for other sensors. Any sensor with Bluetooth enabled can be connected to the phone for standalone use. The manager sensor is the only one capable of forwarding commands to other sensors when multiple sensors are used for data acquisition. Bluetooth sensors will appear as shown in Figure 2.3(b), where the connection name includes the sensor ID number and the serial number.

2. **Activity Log:** To see a log of the most recent sensor messages.
3. **Radio Test:** You can perform a radio test to evaluate the communication link quality between the sensor manager and the rest of the sensors. The default

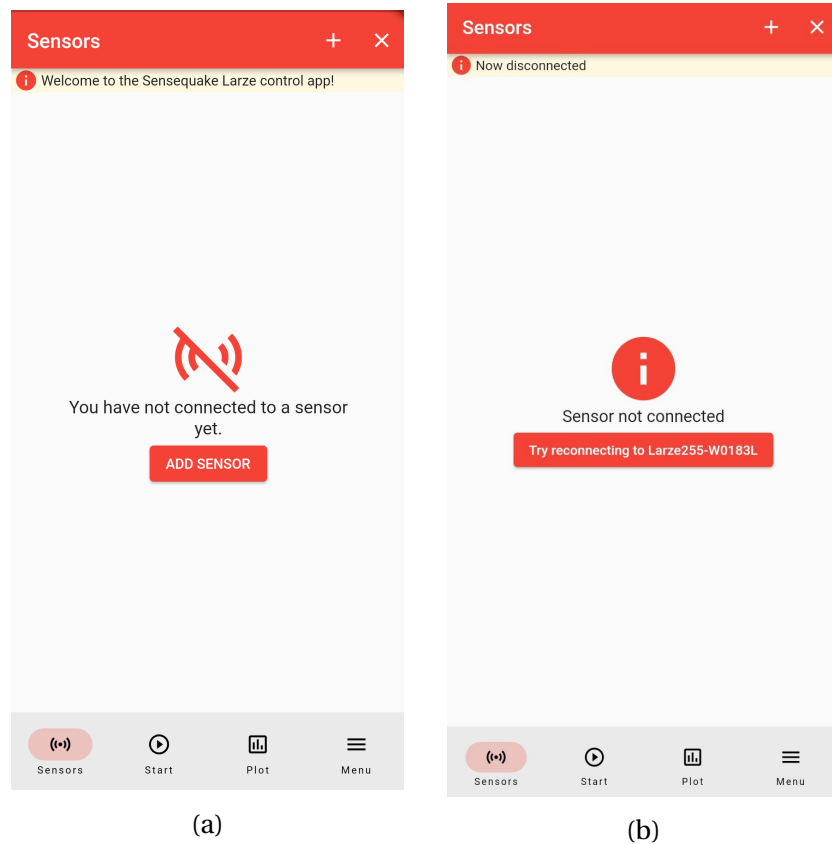


Figure 2.2: (a) The main menu in the Larzé phone app before addition of any sensors. (b) The main menu after memorizing a sensor.

number of packets transmitted is 500. You can reduce this number for faster operation.

Figure 2.4 shows the results obtained from a radio test. The figure illustrates the packet transmission success rate as a percentage, as well as the signal strength in dBm. Radio RSSI (Received Signal Strength Indicator) measures the power level of the received signal. RSSI values are negative, with a value closer to zero indicating a stronger signal and a more positive value (less negative) indicating a weaker signal. For example, an RSSI of -30 dBm indicates a strong signal, while an RSSI of -90 dBm indicates a weak signal, which may result in packet drops. It is recommended to bring the sensors closer together or elevate the antennas to improve signal quality.

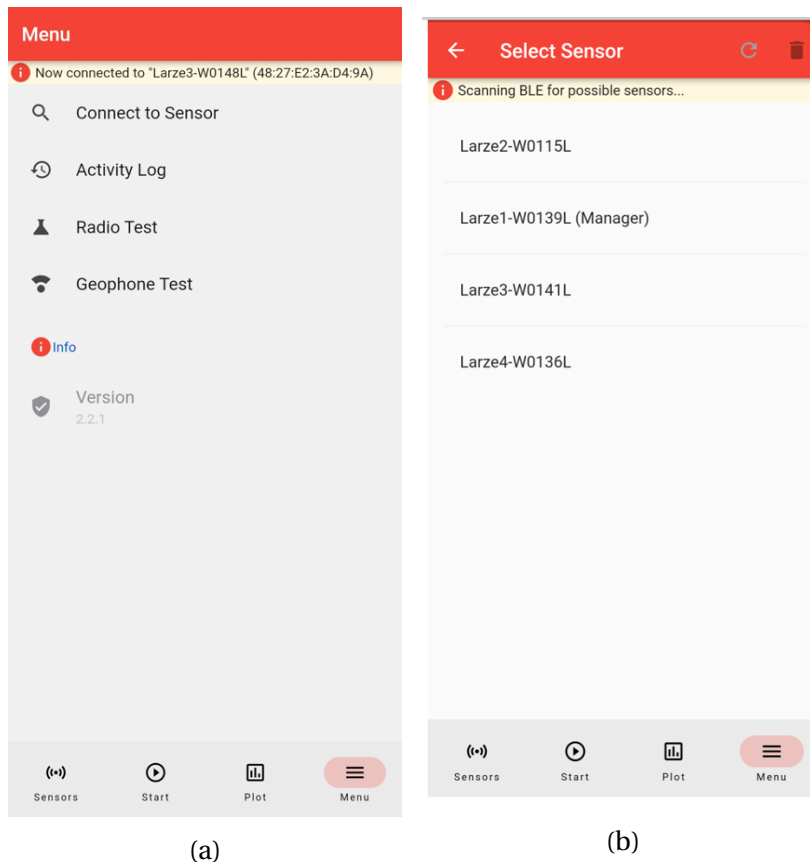


Figure 2.3: The main menu and the sensors page in the Larzé phone app.

4. **Geophone Test:** To perform an electrical test on the geophones.

2.2 Sensors

You can add sensors to the data acquisition network by using the + sign at the top of the bar (Figure 2.5). For instance, to add sensors 2, 3, and 5, enter 2,3,5. This will create a sensor network including sensors 1, 2, 3, and 5.

The information on this page refreshes every 10 seconds by default, though this interval can be adjusted in the sensor configuration file. Here, you can review the sensor battery level, serial number, firmware version, and the time of the last status update. If the radio connection between an expected sensor and the man-

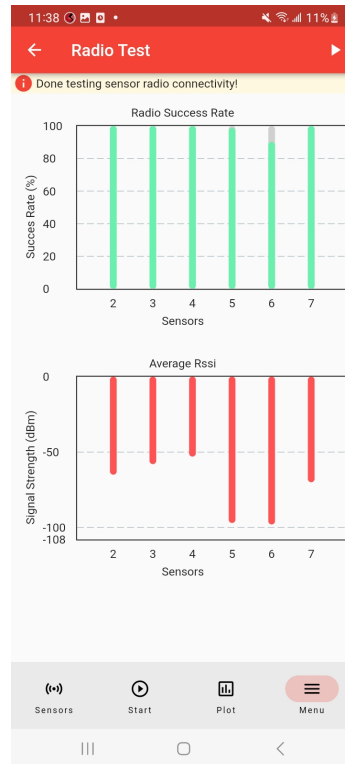


Figure 2.4: The radio test results.

ager is lost, it will appear as a gray box. Before starting data collection, ensure that all sensors are displayed as green boxes with a good connection to the manager.

2.3 Start

The start menu (Figure 2.6) is divided into the following two tabs:

1. **Main** you can specify the project name, setup number, data recording duration, test start delay and sampling frequency in Hertz.

The start delay is particularly useful for testing large structures, as it allows sensors to be synchronized while in close proximity before being moved to their designated positions. Sensors will begin recording once the delay period has elapsed.

2. **Settings**: The following settings should be specified:

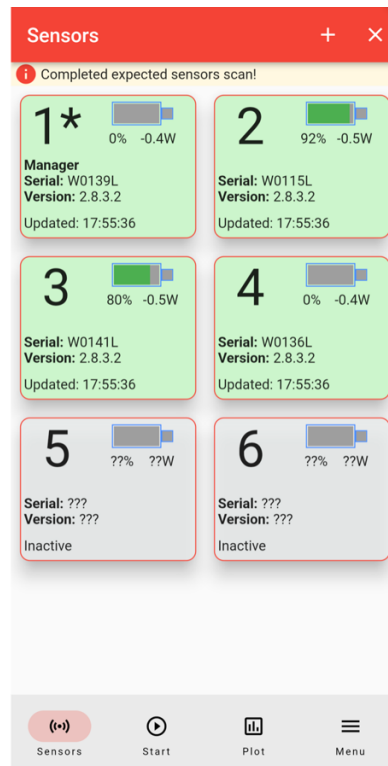


Figure 2.5: The expected sensors in the Larzé phone app.

- **Sync Mode:** choose between Radio (default), GPS, or Periodic GPS. For reliable radio synchronization, external antennas should be connected. GPS synchronization requires a clear view of the sky and may take longer to establish, but it is useful for sensors placed far apart, outside of radio coverage, or when precise timestamps are needed. The absolute GPS time will be added to the recorded files. The Start GPS Now button appears on the Main tab to enable the GPS module before starting a test. When using GPS, the delay field is replaced with a Date field to specify the exact time for starting data recording. The Periodic GPS synchronization option helps correct sensor drift during extended data records lasting several hours.
- **Active Channels:** You can independently select which velocity and acceleration channels are going to be recorded
- **Full Scale Range:** To specify the maximum acceleration that can be measured. The 2g range results in the lowest self noise level.

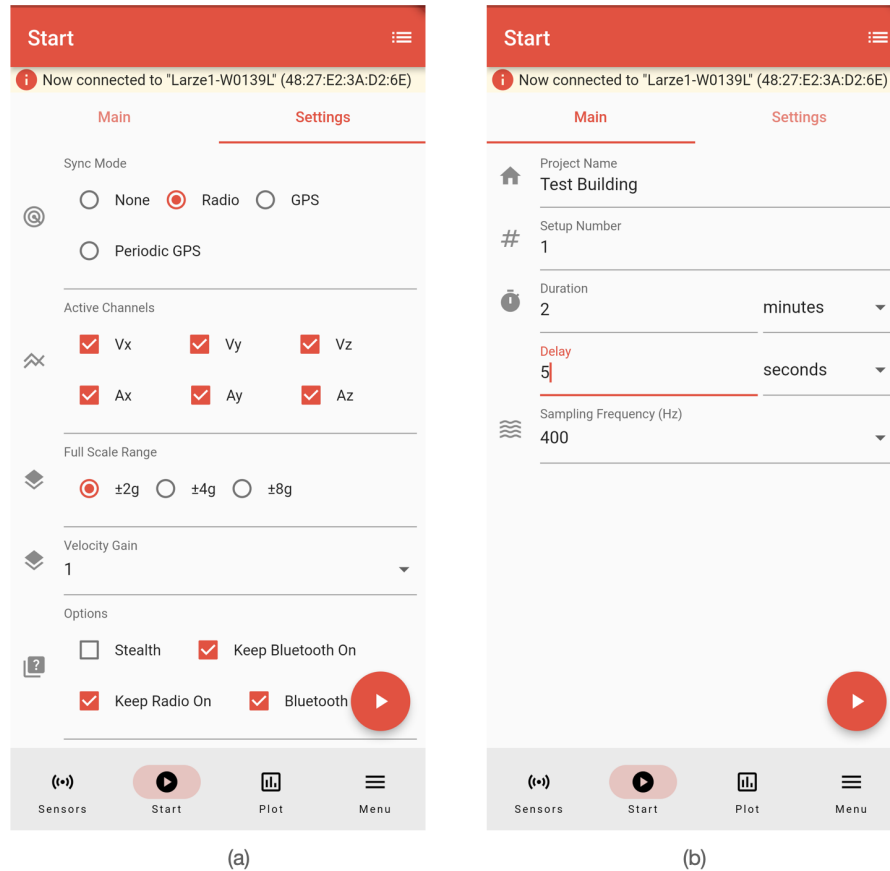
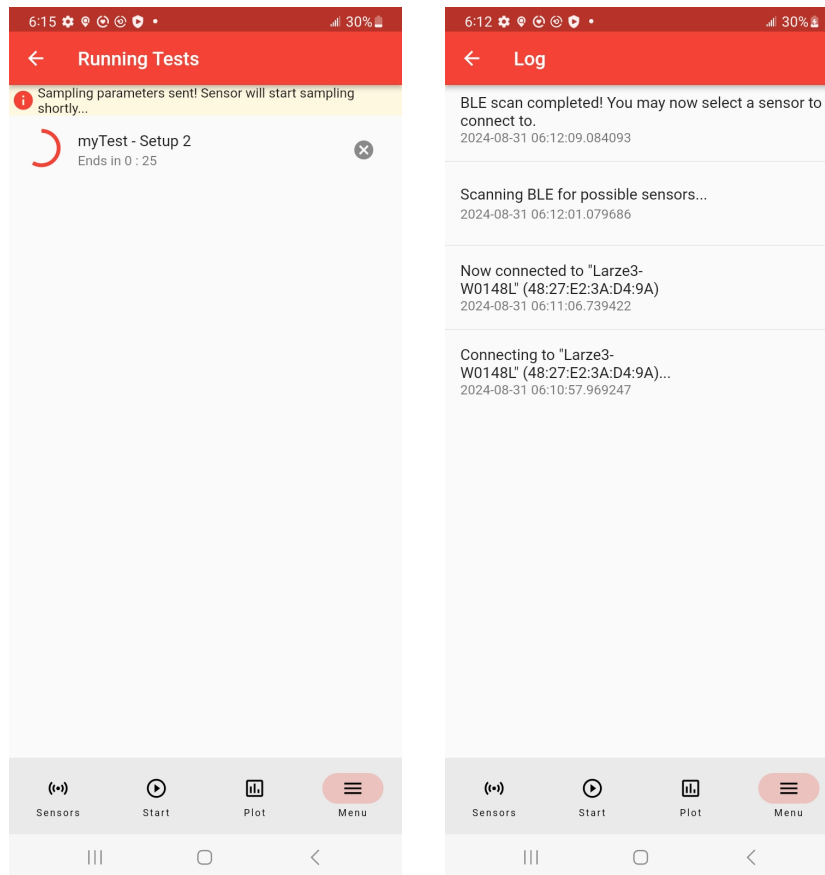


Figure 2.6: The start menu in the Larzé phone app.

- **Velocity Gain:** You can select a gain setting between 1 and 128. A higher gain is useful for measuring very low velocity levels, but be cautious—setting the gain too high may cause channel saturation and clipping
- **Options:** Stealth turns off the LED during data recording. You can choose to keep the Bluetooth and the radio on during the test. Select Bluetooth real time if you would like to see data plots while the test is ongoing.

To monitor the progress of your measurement, tap the icon in the top-right corner of the app (represented by three dots and lines). This will display a list showing both the ongoing measurement progress and the completed tests (Fig-



(a) Measurement progress, (b) logs.

ure 2.7(a)). If you need to cancel a test in progress, simply click the cross button, provided that the sensor is connected to the app via Bluetooth.

The most recent operation log entry is displayed within the yellow ribbon at the top of the screen. To view additional logs, tap on the yellow area (see Figure 2.7(b)).

2.4 Plot

You can visualize the vibration data in the plot menu (Figure 2.8). The following options are available:

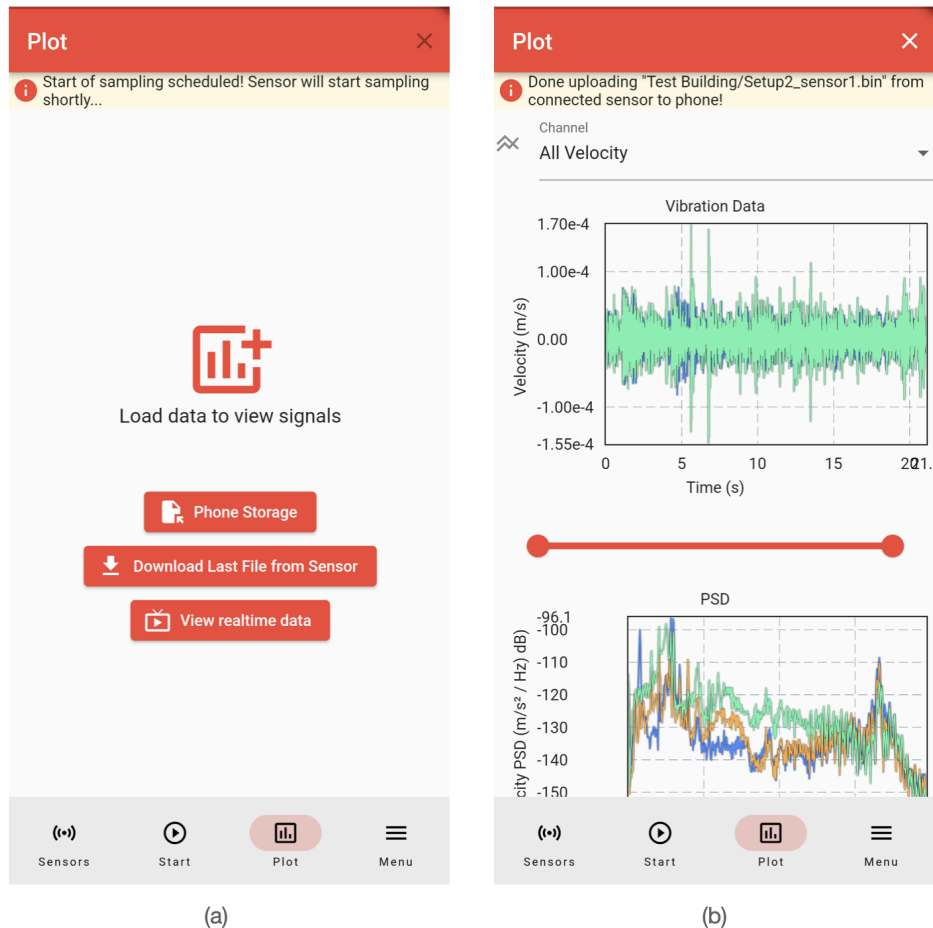


Figure 2.8: The plot menu in the Larzé phone app.

- **Phone Storage:** Load a binary file previously recorded by a sensor from your phone storage.
- **Download Last File From Sensor:** You can download the last file recorded from a sensor connected to the phone via Bluetooth.
- **View Real-time Data:** Plot incoming vibration data in real-time. The data will be streaming from the sensor to the phone while also being saved in the sensor internal memory.

LARZÉ SENSORS

Larzé Vibration System is mainly designed for ambient vibration testing of civil structures. The sensors are very low-noise and extremely sensitive making them capable of receiving the smallest vibrations without the need to use any external force to shake the structure. Multiple sensors can be deployed on structure to collect velocity and acceleration data in a time synchronous manner. The data records can be used for modal identification, damage detection, or seismic response predictions.

Fig. 3.1 shows the Larzé sensor attached to a leveling plate. The sensor dimensions are $L=120$ mm, $W=80$ mm, $H=90$ mm, and it weighs 700 grams.

There is an RGB LED located on top of the enclosure providing the following indications:

- **ON:** Flashing green indicates that the sensor is powered on and functioning correctly.
- **LOW BATTERY:** Flashing red indicates low battery. Charging the battery is recommended.
- **DATA ACQ.:** Solid blue indicates that the start command has been received. Fast flashing blue signals that data acquisition is in progress. The blue LED will stop flashing once the test is completed.
- **GPS:** the yellow LED flashes once every second when GPS signals are received. The sensor must have a clear view of the sky to receive GPS signals. Faster flashing if the yellow LED means that GPS connection is not established yet.
- **WIFI:** the magenta LED shows Wi-Fi activity.



Figure 3.1: The Larzé sensor.

- **USB:** the cyan LED flashes if the sensor is connected to a computer via a micro-USB cable.

The following ports are located on the left side of the sensors (Fig. 3.1):

- **USB:** micro-USB port to connect the sensor to a PC or to charge the battery
- **RADIO:** female SMA connector for attaching an antenna, required for all multi-sensor tests.

Each sensor is labeled with an ID number. The manager sensor, with an ID of 1, connects to the phone app to receive user commands. The manager sensor configures the rest of the sensors using a star radio network. This radio network synchronizes the sensor network with a precision of approximately $5 \mu\text{s}$.

3.1 Sensing Elements

The Larzé sensor consists of a triaxial velocimeter and a triaxial MEMS accelerometer. Please refer to the Appendix section 4.1 for specifications.

The analog front-end has been carefully designed to ensure ultra low noise performance. Independent high performance 24-bit Analog-to-digital converters (ADCs) are used on each channel to resolve very small ambient vibration levels. The user can select between velocity and acceleration or enable both. The accelerometer measurement range can be selected between $\pm 2g$, $\pm 4g$, and $\pm 8g$.

Larzé sensors need to be placed on flat surfaces for accurate velocity measurements. It is recommended to use a bubble level to ensure that the surface is flat. A leveling plate can be used temporarily or permanently attached with screws to the bottom of the sensor units.

3.2 Test Procedure

Please follow the following steps to start a vibration test. You can refer to Chapter 2 for details about using the phone app.

1. Switch on the sensors and verify that the LED is flashing green and that the sensors are not low on battery.
2. Connect the phone to the manager sensor via Bluetooth.
3. Attach the antenna to the radio port if you are using multiple sensors.
4. Add the expected sensors to the list if they are not already present. They should appear in green boxes if the sensors are able to connect to the manager via radio.
5. Select your desired test parameters in the phone app, then press the play button to start recording data. The sensors data acquisition LED will turn on after the start command is sent and will flash while data is being recorded.
6. The sensors will restart after the completion of a test, and the blue LED will stop flashing.
7. The recorded data can be retrieved by connecting the sensor to a PC via the micro-USB port.

Note: Always eject the sensor from your computer before unplugging the USB cable to avoid SD card corruption.

Note: The radio connection is essential for sending commands to the sensors. Therefore, ensure that the sensor units are within an acceptable distance of each other, depending on the radio environment. You can use the radio test command to verify that the signal reception is adequate. Additionally, the phone must be within Bluetooth range of the device to successfully send commands.

Note: The Bluetooth and radio communications are turned off during the measurement to save on battery. It is not possible to interact with the sensor at this time. You can keep the Bluetooth active during the test to enable real-time streaming of the measured data to the phone.

3.3 Data Records

The measurement results are digitized and saved on sensor internal flash memory. The 16 GB storage contains folders named after project names and files named as “Setupx_sensory.txt”, where x is a number indicating the setup number and y is the sensor ID.

Multi-setup tests are commonly used in ambient vibration studies. In these tests, one or two sensors, designated as references, remain in fixed positions, while the remaining sensors are moved to various locations of interest throughout the structure. The data collected from these sensors is combined during post-processing to facilitate modal identification.

Data records can be accessed using a micro-USB connection to a PC to download, or delete files. The sensor will open up a new disk drive on your PC a few seconds after USB connection. The LED indicates a USB connection with a cyan color.

Figure 3.2 shows a sample file recorded by a sensor. The file name consists of the setup number and the sensor number. The header shows some of the measurement parameters such as the test duration and the sampling frequency. The sensor data is encoded in binary format, which could be decoded using Sensequake’s dashboard/data converters.

3.4 Sensor Configuration

Sensor parameters can be configured by opening `editConfig.html` in the sensor’s file system using a browser. First, navigate to `config.txt` in the same file

CHAPTER 4

APPENDIX

4.1 SPECIFICATIONS

Table 4.1 shows the specifications of the Larzé Vibration Monitoring System.

Table 4.1: Specifications

Sensors	
Triaxial velocimeter	120 mm/s
Triaxial accelerometer	$\pm 2g$, $\pm 4g$, and $\pm 8g$
Digital Conversion	
Number of differential channels	3
ADC resolution	24
Sampling rates	10 Hz to 4800 Hz
Dynamic range	148 dB at 16 Hz and 136 dB at 400 Hz sampling
Built-in analog and digital anti-aliasing filters	
Self-noise Density Estimate	
Accelerometers	$25 \mu g/\sqrt{Hz}$
Velocimeters	$20 ng/\sqrt{Hz}$ between 1 to 20 Hz
Synchronization	
Using 2.4 GHz radio	$5 \mu s$
Using GPS	under $1 \mu s$
Communication	
BLE (Bluetooth low energy)	
Sub-GHz radio	

USB for data transfer and firmware updates	
Storage	
Capacity	16 GB
File System	FAT32
LEDs	
Power	
Low battery/errors	
Data acquisition	
Wi-Fi connection	
GPS signal reception	
Phone Application	
Operating System	Android
Power Supply	
Internal lithium battery	about 6 days of operation
Micro-USB port for charging	
Solar charging (optional)	
Connectors	
Radio antenna	SMA
Charge/Data port	Micro-USB
Dimensions	120 x 90x 80 mm
Weight	700 g
Splash proof body	
Temperature Range	-30 to 50° C

4.2 FCC Statements

This device complies with part 15 of the FCC rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Changes or modifications to this unit not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

NOTE: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help

4.3 IC Statements

This device contains licence-exempt transmitter(s)/receiver(s) that comply with Innovation, Science and Economic Development Canada's licence-exempt RSS(s). Operation is subject to the following two conditions: (1) This device may not cause interference. (2) This device must accept any interference, including interference that may cause undesired operation of the device.

L'émetteur/récepteur exempt de licence contenu dans le présent appareil est conforme aux CNR d'Innovation, Sciences et Développement économique Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes : (1) L'appareil ne doit pas produire de brouillage; (2) L'

l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

RF Exposure Warning Statements: This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. This equipment shall be installed and operated with minimum distance 20cm between the radiator and body.

Radio Frequency Exposure Statement for IC: The device has been evaluated to meet general RF exposure requirements. The device can be used in mobile exposure conditions. The min separation distance is 20 cm.

Déclaration d'exposition aux radiofréquences pour IC: L'appareil a été évalué pour répondre aux exigences générales en matière d'exposition aux RF. L'appareil peut être utilisé dans des conditions d'exposition mobiles. La distance de séparation minimale est de 20 cm.